

Q1 Please see Printed Pg 1 for Working & calculations

Case	V_{gs}	V_{ds}	V_{gt}
1	0.25	0.8	-0.1
2	0.55	0.1	0.2
3	0.7	0.4	0.35
4	0.9	0.4	0.55
5	1	0.8	0.65

$$V_t = 0.35$$

$$V_{dd} = 1$$

$$V_{gt} = V_{gs} - V_{ds}$$

Q1a Long-channel model $\rightarrow V_{dsat} = V_{gt}$.

Case	V_{gt}	V_{ds}	V_{dsat}	Region of operation.
1	-0.1	0.8	-0.1	OFF. $V_{gt} < 0$.
2	0.2	0.1	0.2	Linear. $V_{gt} > 0, V_{ds} < V_{dsat}$.
3	0.35	0.4	0.35	Saturated. $V_{gt} > 0, V_{ds} > V_{dsat}$.
4	0.55	0.4	0.55	Linear. $V_{gt} > 0, V_{ds} < V_{dsat}$.
5	0.65	0.8	0.65	Saturated. $V_{gt} > 0, V_{ds} > V_{dsat}$.

Q1b Short-channel model $\rightarrow V_{dsat} = \min(V_{gt}, 0.45)$

case	V_{gt}	V_{ds}	V_{dsat}	Region of operation.
1	-0.1	0.8	-0.1	Off. $V_{gt} < 0$.
2	0.2	0.1	0.2	Linear. $V_{gt} > 0, V_{ds} < V_{dsat}$.
3	0.35	0.4	0.35	Saturated. $V_{gt} > 0, V_{ds} > V_{dsat}$.
4	0.55	0.4	0.45	Linear. $V_{gt} > 0, V_{ds} < V_{dsat}$.
5	0.65	0.8	0.45	Saturated. $V_{gt} > 0, V_{ds} > V_{dsat}$.

Q1c For the information specified, there are no cases where the long-channel & short channel models give different operating-region classifications. There are no disagreements to explain.

Q1d Early velocity saturation changes the intuition that drain current increases quadratically with V_{gs} because while in long-channel transistors where saturation happens through pinch-off, short-channel transistors are saturated far earlier (normally due to pinch off but now due to induced electric fields causing saturation velocity v_{sat}).

Case	Vgs (V)	Vds (V)	Vgt (V)
1	0.25	0.8	-0.1
2	0.55	0.1	0.2
3	0.7	0.4	0.35
4	0.9	0.4	0.55
5	1	0.8	0.65

Vt (V)	0.35
Vdd (V)	1

Long Channel Model		Vdsat = Vgt		
Case	Vgt	Vds (V)	Vdsat	Reg. of. Op.
1	-0.1	0.8	-0.1	Vgt<0, OFF
2	0.2	0.1	0.2	Vgt>0, Vds<Vdsat. LINEAR
3	0.35	0.4	0.35	Vgt>0, Vds>Vdsat. SATURATED
4	0.55	0.4	0.55	Vgt>0, Vds<Vdsat. LINEAR
5	0.65	0.8	0.65	Vgt>0, Vds>Vdsat. SATURATED

Short-Channel Model		Vdsat = MIN(Vgt, 0.45V)		
Case	Vgt	Vds (V)	Vdsat	Reg. of. Op.
1	-0.1	0.8	-0.1	Vgt<0, OFF
2	0.2	0.1	0.2	Vgt>0, Vds<Vdsat. LINEAR
3	0.35	0.4	0.35	Vgt>0, Vds>Vdsat. SATURATED
4	0.55	0.4	0.45	Vgt>0, Vds<Vdsat. LINEAR
5	0.65	0.8	0.45	Vgt>0, Vds>Vdsat. SATURATED

Page 1 Working
for Q1.

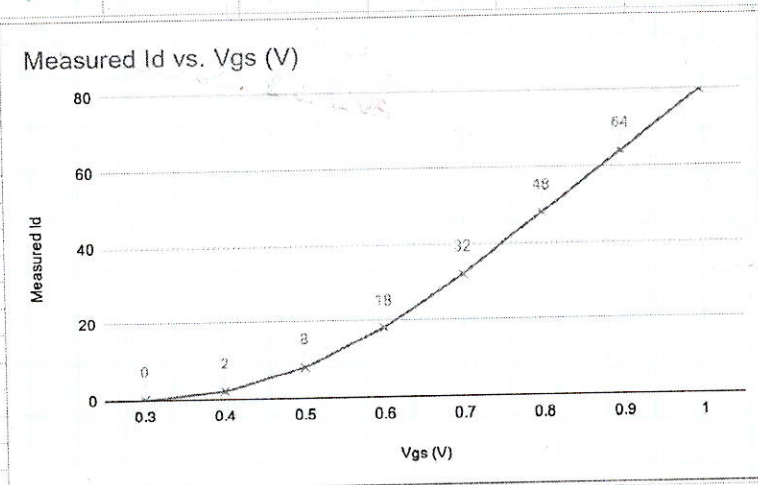
Q1D, cont) ~~at~~ for the carrier electrons far before any pinch-off can start to affect the drain current.

1e) This is not true: If V_{gs} is less than V_{thres} , despite any large V_{ds} value, the gate will remain OFF and therefore not saturated.

Example: Case 1. V_{ds} is 0.8 and yet the gate remains off and not saturated.

2a) After graphing V_{gs} and I_D , the V_T threshold is where I_D first equals 0. This is ~~at~~ 0.3 V.

Vds (V)		1
Vgs (V)	Measured Id	
0.3	0	
0.4	2	
0.5	8	
0.6	18	
0.7	32	
0.8	48	
0.9	64	
1	80	



(doesn't require a graph but I liked the visual reference for future Question 2 parts.)

2b

Computed K at $V_{GS}=0.6$

↳ 200.

Predicted I_D at $V_{GS}=1$

↳ 98.

Actual I_D at $V_{GS}=1$

↳ 80.

Predicted > Actual.

Estimated V_t (V):	0.3	$I_D = K(V_{GS}-V_t)^2$ $K = I_D/(V_{GS}-V_t)^2$	
V_{GS}	Estimated V_t	I_D	K
0.6	0.3	18	200
1	0.3	98	200
1	0.3	80	163.2653061

Bold=Calculated

< using calculated K

< Actual values

Computed K : 200

Predicted I_D at $V_{GS}=1$: 98

Actual I_D at $V_{GS}=1$: 80

QUADRATIC MODEL

Prediction I_D is higher than measured value, and Prediction K is also higher than measured value.

2c

Computed K_v at $V_{GS}=0.8$

↳ 96.

Predicted I_D at $V_{GS}=1$

↳ 67.2

Actual I_D at $V_{GS}=1$

↳ 80.

Predicted < actual.

Estimated V_t (V):	0.3	$I_D = K_v(V_{GS}-V_t)$ $K_v = I_D/(V_{GS}-V_t)$	
V_{GS}	Estimated V_t	I_D	K
0.8	0.3	48	96
1	0.3	67.2	96
1	0.3	80	114.2857143

Bold=Calculated

< using calculated K

< Actual values

Computed K : 96

Predicted I_D at $V_{GS}=1$: 67.2

Actual I_D at $V_{GS}=1$: 80

LINEAR MODEL

Prediction I_D is lower than measured value, and Prediction K is also lower than measured value.

2d

Comparing the two models from 2B & 2C.

Quadratic model:
Prediction I_D is higher than measured value, and Prediction K is also higher than measured value.
Linear model:
Prediction I_D is lower than measured value, and Prediction K is also lower than measured value.

A linear model fits better as saturation is reached.

V_{GS}	I_D
0.3	0
0.4	2
0.5	8
0.6	18
0.7	32
0.8	48
0.9	64
1.0	80

$\begin{matrix} +2 \\ +6 \\ +10 \\ +14 \\ +16 \\ +16 \end{matrix}$

quadratic @ lower V_{GS}

linear @ higher V_{GS} .

(2e) estimated quadratic relationship:

$$I_D = K(V_{GT} - V_T)^2 \rightarrow I_D \propto V^2.$$

↑
Proportional
to

Propagation Delay (Time units) = τ tan?

~~$\tau = \frac{C \cdot \Delta V}{I}$~~

$\tau = \frac{C \cdot \Delta V}{I}$

if $I \propto V^2$, as V improves, I improves
By far more, so τ is not as big as
the developer expects.

~~MSA~~

$V_m = \frac{V_{DD}}{2 + r}$

$$(3a) \quad I_{DN} = I_{DP} \quad V_{TN} = |V_{TP}| = 0.3V. \quad V_{DD} = 1V$$

$$K_n W_N (V_M - V_{TN}) = K_p W_{BP} (V_{DD} - V_M - |V_{TP}|)$$

$$\frac{K_N}{K_P} (V_M - V_{TN}) = \frac{W_P}{W_N} (V_{DD} - V_M - |V_{TP}|)$$

$$2(V_M - V_{TN}) = r(V_{DD} - V_M - |V_{TP}|)$$

$$2V_M - 2(0.3) = rV_{DD} - rV_M - 0.3r$$

$$2V_M + rV_M = r(1) - 0.3r + 0.6$$

$$(2+r)V_M = 0.7r + 0.6$$

$$V_M = \frac{0.7r + 0.6}{2+r}$$

$$(3b) \quad V_M = V_{DD}/2 = 1/2$$

$$0.5 = \frac{0.7r + 0.6}{2+r}$$

$$(2+r)(0.5) = 0.7r + 0.6$$

$$1 + \frac{r}{2} = 0.7r + 0.6$$

$$0.4 = 0.2r$$

$$4 = 2r$$

$$2 = r$$

3c)

$$V_M = \frac{0.7r + 0.6}{2+r}$$

$$\frac{W_P}{W_N} = R$$

$$r=1, \quad V_M = \frac{0.7+0.6}{3} = \frac{1.3}{3} = \frac{13}{30} V$$

$$r=2, \quad V_M = \frac{1.4+0.6}{4} = \frac{2}{4} = \frac{1}{2} V$$

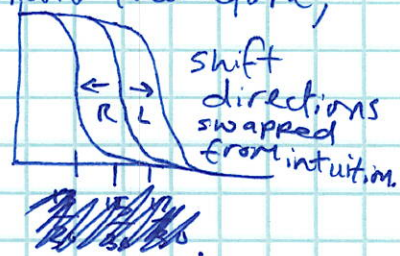
← 15/30 for ease of comparison.

$$r=4, \quad V_M = \frac{2.8+0.6}{2+4} = \frac{3.4}{6} = \frac{34}{60} = \frac{17}{30} V$$

Increasing R , $\frac{W_P}{W_N}$, makes the PMOS side of the

inverter stronger. This means that the transfer curve towards the PMOS side, where the output approaches zero, which is off to the ~~right~~ left now. therefore, shift right.

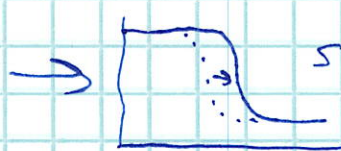
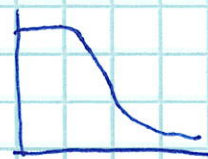
electronics.stackexchange.com/questions/593213/vtc-curve-and-the-delay-time-of-cmos-inverter



~~3d)~~

3e)

If NMOS is stronger than PMOS*, it stays dominant for longer, so is shifted left.



shift left.

*than intended,

3d)

gain ~~amplifier~~ magnitude = amount the inverter amplifies the signal away from V_M . $N_{ML} = V_{IL} - V_{OL}$, $N_{MH} = V_{IH} - V_{OH}$. To improve N_{ML} , where the input-low noise affects, choose a higher V_M to increase V_{IL} .

4a

Vdd (V)

1

 $I_{dn} = I_{dp}$

Vtn (V)

0.3

 $K_n/K_p = 2$

|Vtp| (V)

0.3

 $W_p/W_n = r$

$$I_{dn} = k_n W_n (V_m - V_{tn})$$

$$I_{dp} = K_p W_p (V_{dd} - V_m - |V_{tp}|)$$

$$K_n W_n (V_m - V_{tn}) = K_p W_p (V_{dd} - V_m - |V_{tp}|)$$

$$K_n W_n (V_m - V_{tn}) / K_p = W_p (V_{dd} - V_m - |V_{tp}|)$$

$$2 W_n (V_m - V_{tn}) = W_p (V_{dd} - V_m - |V_{tp}|)$$

$$2 W_n (V_m - 0.3) = W_p (1 - V_m - 0.3)$$

$$2 (V_m - 0.3) = W_p (1 - V_m - 0.3) / W_n$$

$$2 V_m - 0.6 = r (0.7 - V_m)$$

$$2 V_m - 0.6 = 0.7r - r V_m$$

$$2 V_m + r V_m = 0.7r + 0.6$$

$$(2+r) V_m = 0.7r + 0.6$$

$$V_m = (0.7r + 0.6) / (2+r)$$

Design	Wn	Wp	r	Vm
A	1	1	1	0.4333333333
B	1	2	2	0.5
C	1	3	3	0.54

$$\frac{K_n}{K_p} = 2$$

4b

Vdd (V)

1

 $I_{dn} = I_{dp}$

Vtn (V)

0.3

 $K_n/K_p = 1.6$

|Vtp| (V)

0.3

 $W_p/W_n = r$

$$I_{dn} = k_n W_n (V_m - V_{tn})$$

$$I_{dp} = K_p W_p (V_{dd} - V_m - |V_{tp}|)$$

$$K_n W_n (V_m - V_{tn}) = K_p W_p (V_{dd} - V_m - |V_{tp}|)$$

$$K_n W_n (V_m - V_{tn}) / K_p = W_p (V_{dd} - V_m - |V_{tp}|)$$

$$1.6 W_n (V_m - V_{tn}) = W_p (V_{dd} - V_m - |V_{tp}|)$$

$$1.6 W_n (V_m - 0.3) = W_p (1 - V_m - 0.3)$$

$$1.6 (V_m - 0.3) = W_p (1 - V_m - 0.3) / W_n$$

$$1.6 V_m - 0.6 = r (0.7 - V_m)$$

$$1.6 V_m - 0.6 = 0.7r - r V_m$$

$$1.6 V_m + r V_m = 0.7r + 0.6$$

$$(1.6+r) V_m = 0.7r + 0.6$$

$$V_m = (0.7r + 0.6) / (1.6+r)$$

$$\frac{K_n}{K_p} = 1.6$$

changed derivation
to slow working

Design	Wn	Wp	r	Vm
A	1	1	1	0.5
B	1	2	2	0.5555555556
C	1	3	3	0.5869565217

$$\frac{K_n}{K_p} = 1.6$$

derivation
typed on
google
sheets
for legibility

4c

Vdd (V)	1	$I_{dn} = I_{dp}$
Vtn (V)	0.3	$K_n/K_p = 2.6$
Vtp (V)	0.3	$W_p/W_n = r$
$I_{dn} = k_n W_n (V_m - V_{tn})$		$I_{dp} = K_p W_p (V_{dd} - V_m - V_{tp})$

$$K_n W_n (V_m - V_{tn}) = K_p W_p (V_{dd} - V_m - |V_{tp}|)$$

$$K_n W_n (V_m - V_{tn}) / K_p = W_p (V_{dd} - V_m - |V_{tp}|)$$

$$2.6 W_n (V_m - V_{tn}) = W_p (V_{dd} - V_m - |V_{tp}|)$$

$$2.6 W_n (V_m - 0.3) = W_p (1 - V_m - 0.3)$$

$$2.6 (V_m - 0.3) = W_p (1 - V_m - 0.3) / W_n$$

$$2.6 V_m - 0.6 = r(0.7 - V_m)$$

$$2.6 V_m - 0.6 = 0.7r - rV_m$$

$$2.6 V_m + rV_m = 0.7r + 0.6$$

$$(2.6 + r)V_m = 0.7r + 0.6$$

$$V_m = (0.7r + 0.6) / (2.6 + r)$$

$$\frac{K_n}{K_p} = 2.6$$

← for all taped-in grids, used formula to calculate on google sheets to avoid copying errors

Design	Wn	Wp	r	Vm
A	1	1	1	0.3611111111
B	1	2	2	0.4347826087
C	1	3	3	0.4821428571

4d

Robustness.

$$\Delta V_m = V_{M \text{ MAX}} - V_{M \text{ MIN.}}$$

Design	Nominal	Slow Nmos Fast Pmos	Fast Nmos Slow Pmos	Vm Spread
A	0.4333333333	0.5	0.3611111111	0.1388888889
B	0.5	0.5555555556	0.4347826087	0.1207729469
C	0.54	0.5869565217	0.4821428571	0.1048136646

Design C has the smallest Vm spread

Vdd/2: 0.5

Design B is closest to Vdd/2 under nominal conditions.

4e

For a noisy environment, need to maximise noise margins to ensure a larger range of values that aren't too close to point where $V_{in} = V_{out}$.

low Vm spread for higher reliability when noisy. despite how heavily Design C is weighted area-wise, due to the size of the PMOS, the size was not mentioned as a limiting factor and the increased size makes it more resistant to manufacturing variability ("sensitivity to process variation") would choose C!

5a) C_{gc} in cutoff, linear, saturation.

"gate channel capacitance ~~is~~ is approximately zero in cutoff" - homework bullet point.

$$C_{gc}(\text{cutoff}) \approx 0.$$

"In linear operation, total gate channel capacitance is $C_{ox}WL$ & is split evenly between source & drain" - HW.

$$C_{ox} = 20 \text{ fF}/\mu\text{m}^2. \quad W = 1 \mu\text{m}. \quad L = 0.1 \mu\text{m}.$$

$$C_{gc}(\text{Linear}) = C_{ox}WL = 20 \times 1 \times 0.1 = 2 \text{ fF}.$$

← split between source & drain equally.

"In saturation, total $[C_{gc}]$ is $\frac{2}{3} \times C_{ox}WL$ and is assigned to the source side." - HW.

$$C_{ox}WL = 2 \text{ fF}. \quad C_{gc} = \frac{2}{3} C_{ox}WL = \frac{4}{3} \text{ fF}.$$

← assigned to source side.

5b)

$$C_{gso} = C_{gdo} = C_o W = 0.25 \times 1 = 0.25.$$

$$\begin{matrix} \swarrow & \swarrow & \uparrow & \uparrow \\ C_s = 0.25 & C_d = 0.25 & C_o & W. \end{matrix}$$



total fringe capacitance $\Rightarrow C_{\text{fringe, total}} = C_{\text{fringe}} W.$

$$C_{\text{fringe}} = 0.2 \text{ fF}. \quad W = 1 \mu\text{m}.$$

$$C_{\text{fringe total}} = 0.4$$

$C_{\text{fringe total}} = C_{gso} + C_{gdo}$? unsure why else it would be calculated.

have to add diffusion capacitance.

$$C_{\text{diff}} = C_j L_s W + C_{jsw} (2L_s + W) + C_{jgate} W.$$

$$= 1 \times 0.15 \times 1 + 0.2 (2 \times 0.15 + 1) + 0.1 \times 1$$

$$= 0.15 + 0.2 (1.3) + 0.1$$

$$= 0.15 + 0.26 + 0.10 = 0.51.$$

← ~~diffusion~~ diffusion capacitance per gate terminal.

$$0.51 \times 2 = 1.02 \text{ fF} : \text{total source \& drain diffusion capacitance.}$$

ASK YANG WU!

5c)

5d)

Miller effect:

$$A_v = -5$$

$$C_{GDO \text{ effective}} \approx C_{GDO} (1 - A_v)$$

$$1 - (-5) = 6$$

$$C_{GDO \text{ effective}} \approx 6 C_{GDO}$$

$$6 \times C_{GDO} = 6 \times 0.25 = 1.5 \text{ pF}$$

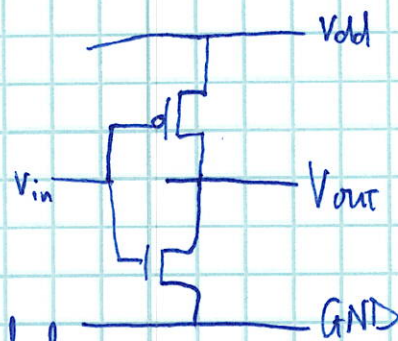
~~Physical Class~~

ASK
YANG
WU!

Is it just that it contributes 1.5 pF to the input capacitance?? And that the effective contribution is 6x as big as the actual C_{GDO} ?

5e)

When the input changes from 0 to 1 or vice versa, the Miller effect is fully ignored by the student. The Miller effect is noticed during the changing of inputs and so by forgetting it the student underestimates in their predicted delay.



It is most noticeable when in VTC transition because that is where A_v is measured and that's where the biggest voltage change is.

AI STATEMENT: I didn't use AI, but I heavily used Yang Wu's teachings in his recitation.